N-Channel 100 V (D-S) MOSFET

PRODUCT SUMMARY		
V_{DS}	100	V
$R_{DS(on)}$ V_{GS} = 10 V	32	mΩ
I_D	9	A
Configuration	Single	

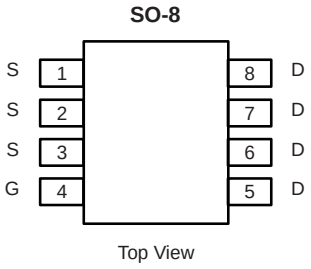
FEATURES

- **Halogen-free According to IEC 61249-2-21 Definition**
- Extremely Low Q_{gd} for Switching Losses
- 100 % R_g Tested
- 100 % Avalanche Tested
- Compliant to RoHS Directive 2002/95/EC

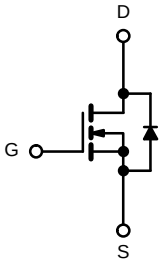
RoHS COMPLIANT
HALOGEN FREE Available

APPLICATIONS

- Primary Side Switch



Top View



N-Channel MOSFET

ABSOLUTE MAXIMUM RATINGS (T_A = 25 °C, unless otherwise noted)				
Parameter		**Symbol**	**Limit**	**Unit**
Drain-Source Voltage		V_{DS}	100	V
Gate-Source Voltage		V_{GS}	± 20	
Continuous Drain Current (T_J = 150 °C)	T_C = 25 °C	I_D	9	A
	T_C = 70 °C		6	
	T_A = 25 °C		6[b, c]	
	T_A = 70 °C		5[b, c]	
Pulsed Drain Current		I_{DM}	40	
Continuous Source-Drain Diode Current	T_C = 25 °C	I_S	7	
	T_A = 25 °C		3.8[b, c]	
Single Pulse Avalanche Current	L = 0.1 mH	I_{AS}	30	
Single Pulse Avalanche Energy		E_{AS}	112	mJ
Maximum Power Dissipation	T_C = 25 °C	P_D	14	W
	T_C = 70 °C		5	
	T_A = 25 °C		4[b, c]	
	T_A = 70 °C		2[b, c]	
Operating Junction and Storage Temperature Range		T_J, T_{stg}	- 55 to 150	°C

THERMAL RESISTANCE RATINGS					
Parameter		**Symbol**	**Typical**	**Maximum**	**Unit**
Maximum Junction-to-Ambient[b, f]	t ≤ 10 s	R_{thJA}	33	40	°C/W
Maximum Junction-to-Foot (Drain)	Steady State	R_{thJF}	17	21	

Notes:
a. Based on T_C = 25 °C.
b. Surface mounted on 1" x 1" FR4 board.
c. t = 10 s.
d. Maximum under steady state conditions is 80 °C/W.

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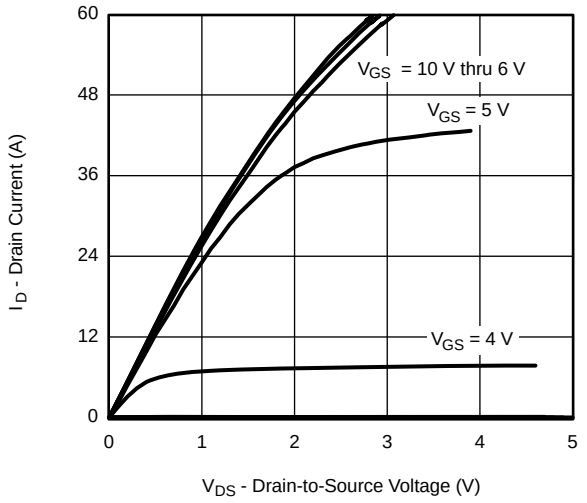
SPECIFICATIONS (T_J = 25 °C, unless otherwise noted)						
Parameter	**Symbol**	**Test Conditions**	**Min.**	**Typ.**	**Max.**	**Unit**
Static						
Drain-Source Breakdown Voltage	V_{DS}	V_{GS} = 0 V, I_D = 250 µA	100			V
V_{DS} Temperature Coefficient	$\Delta V_{DS}/T_J$	I_D = 250 µA		172		mV/°C
$V_{GS(th)}$ Temperature Coefficient	$\Delta V_{GS(th)}/T_J$			- 10		
Gate-Source Threshold Voltage	$V_{GS(th)}$	V_{DS} = V_{GS} , I_D = 250 µA	1.0		3.0	V
Gate-Source Leakage	I_{GSS}	V_{DS} = 0 V, V_{GS} = ± 20 V			± 100	nA
Zero Gate Voltage Drain Current	I_{DSS}	V_{DS} = 100 V, V_{GS} = 0 V			1	µA
		V_{DS} = 100 V, V_{GS} = 0 V, T_J = 55 °C			10	
On-State Drain Current[a]	$I_{D(on)}$	V_{DS} ≥ 10 V, V_{GS} = 10 V	30			A
Drain-Source On-State Resistance[a]	$R_{DS(on)}$	V_{GS} = 10 V, I_D = 5 A		0.032		mΩ
		V_{GS} = 4.5 V, I_D = 5 A		0.033		
Forward Transconductance[a]	g_{fs}	V_{DS} = 15 V, I_D = 5 A		20		S
Dynamic[b]						
Input Capacitance	C_{iss}	V_{DS} = 50 V, V_{GS} = 0 V, f = 1 MHz		1900		pF
Output Capacitance	C_{oss}			150		
Reverse Transfer Capacitance	C_{rss}			50		
Total Gate Charge	Q_g	V_{DS} = 75 V, V_{GS} = 10 V, I_D = 5 A		28.5	43	nC
		V_{DS} = 75 V, V_{GS} = 8 V, I_D = 5 A		23	35	
Gate-Source Charge	Q_{gs}			8		
Gate-Drain Charge	Q_{gd}			6.5		
Gate Resistance	R_g	f = 1 MHz		0.80	1.3	Ω
Turn-on Delay Time	$t_{d(on)}$	V_{DD} = 50 V, R_L = 10 Ω I_D ≅ 5 A, V_{GEN} = 10 V, R_g = 1 Ω		14	21	ns
Rise Time	t_r			12	18	
Turn-Off Delay Time	$t_{d(off)}$			22	33	
Fall Time	t_f			6	10	
Turn-On Delay Time	$t_{d(on)}$	V_{DD} = 50 V, R_L = 10 Ω I_D ≅ 5 A, V_{GEN} = 8 V, R_g = 1 Ω		16	24	
Rise Time	t_r			12	18	
Turn-Off Delay Time	$t_{d(off)}$			20	30	
Fall Time	t_f			7	12	
Drain-Source Body Diode Characteristics						
Continuous Source-Drain Diode Current	I_S	T_C = 25 °C			7.7	A
Pulse Diode Forward Current[a]	I_{SM}				50	
Body Diode Voltage	V_{SD}	I_S = 2.6 A		0.77	1.2	V
Body Diode Reverse Recovery Time	t_{rr}	I_F = 5 A, dI/dt = 100 A/µs, T_J = 25 °C		63	95	ns
Body Diode Reverse Recovery Charge	Q_{rr}			110	165	nC
Reverse Recovery Fall Time	t_a			49		ns
Reverse Recovery Rise Time	t_b			14		

Notes:
a. Pulse test; pulse width ≤ 300 µs, duty cycle ≤ 2 %
a. Guaranteed by design, not subject to production testing.

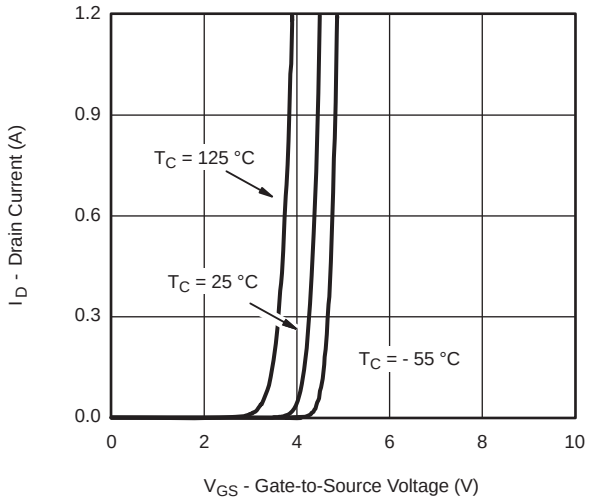
Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

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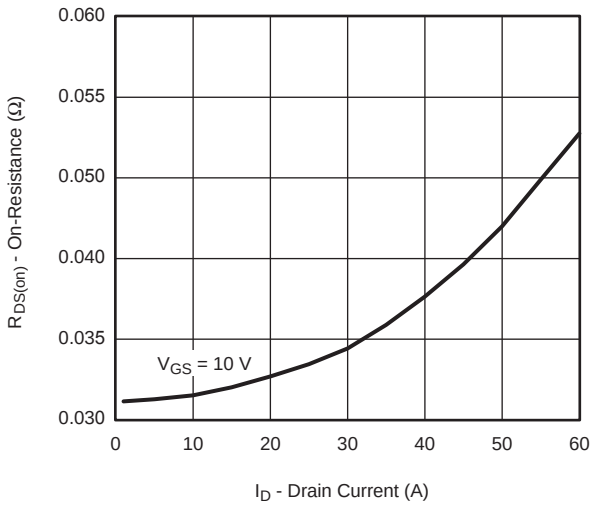
TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



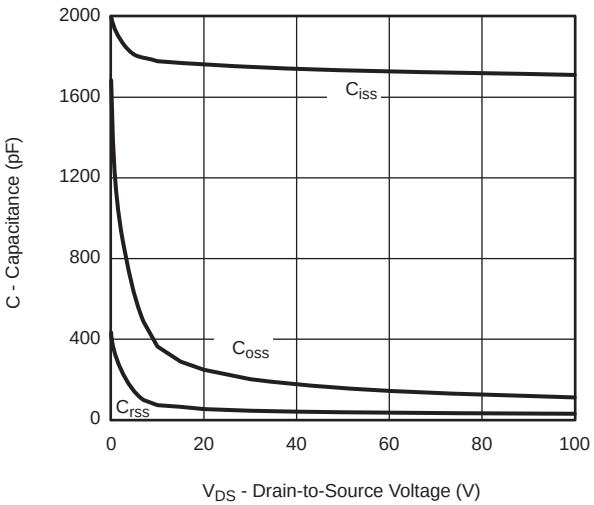
Output Characteristics



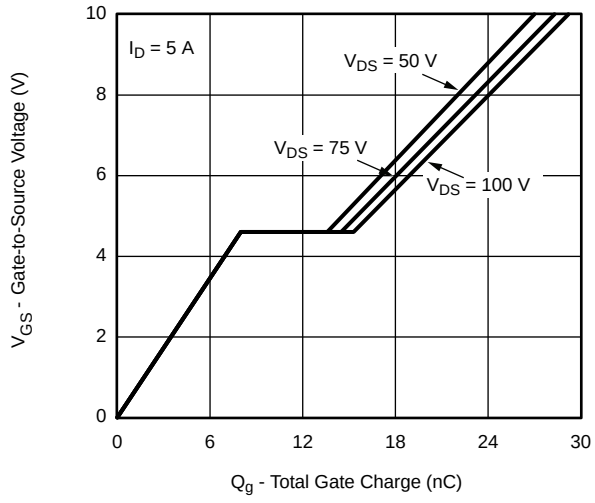
Transfer Characteristics



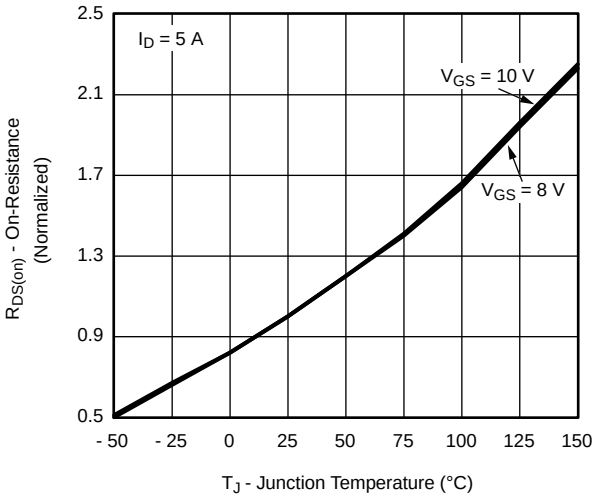
On-Resistance vs. Drain Current and Gate Voltage



Capacitance



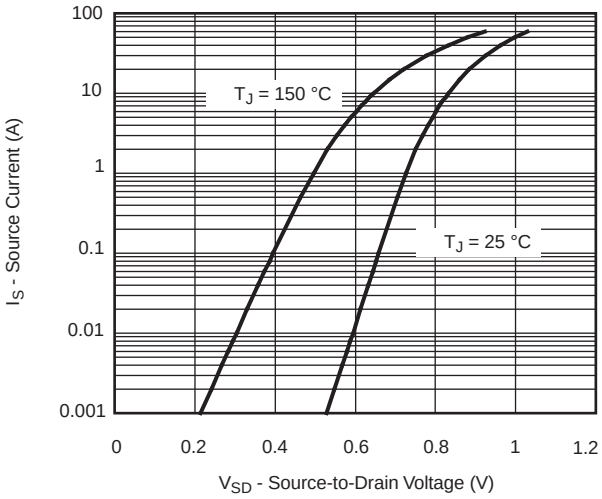
Gate Charge



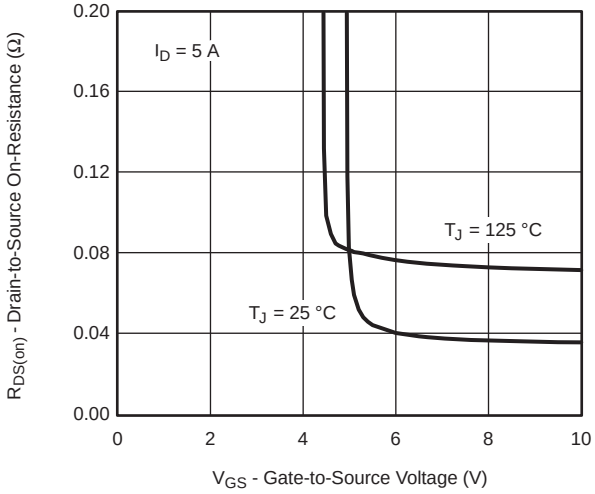
On-Resistance vs. Junction Temperature

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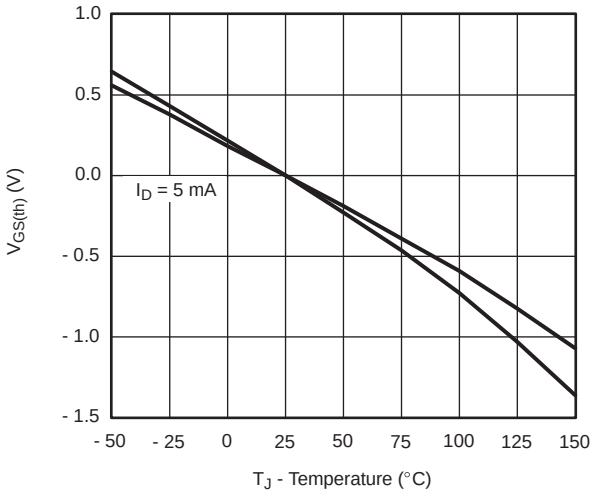
TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



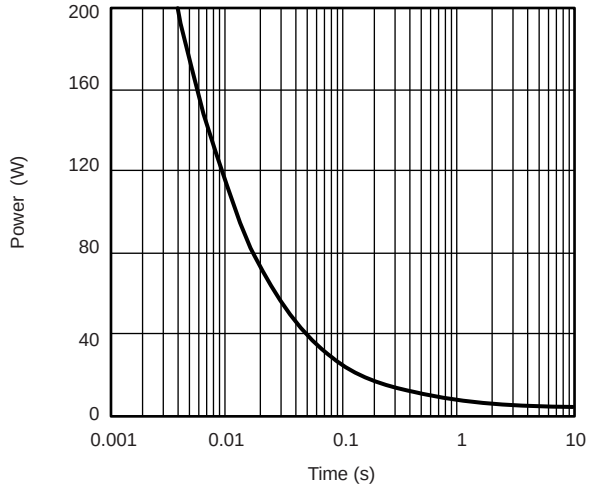
Source-Drain Diode Forward Voltage



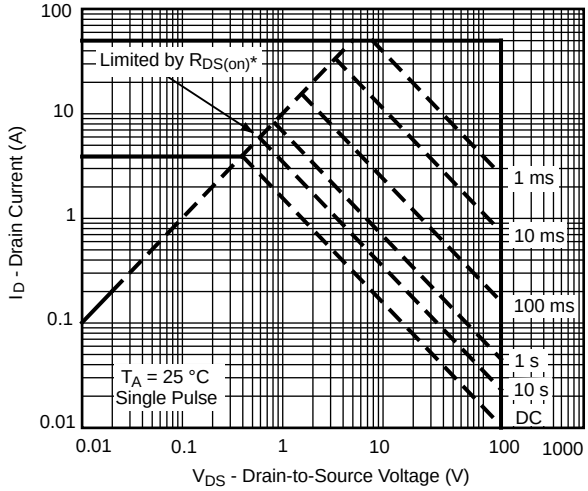
On-Resistance vs. Gate-to-Source Voltage



Threshold Voltage



Single Pulse Power, Junction-to-Ambient

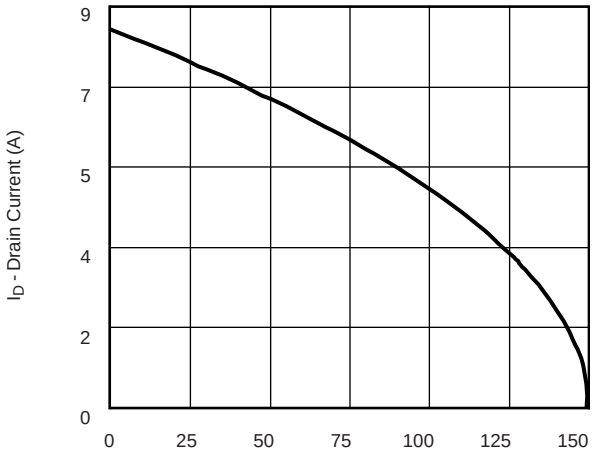


* V_{GS} > minimum V_{GS} at which $R_{DS(on)}$ is specified

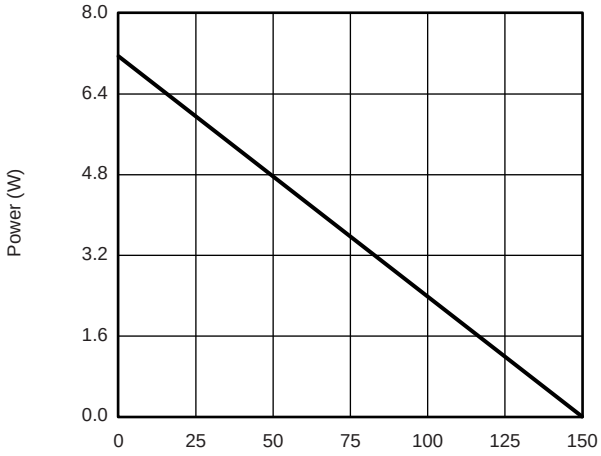
Safe Operating Area, Junction-to-Ambient

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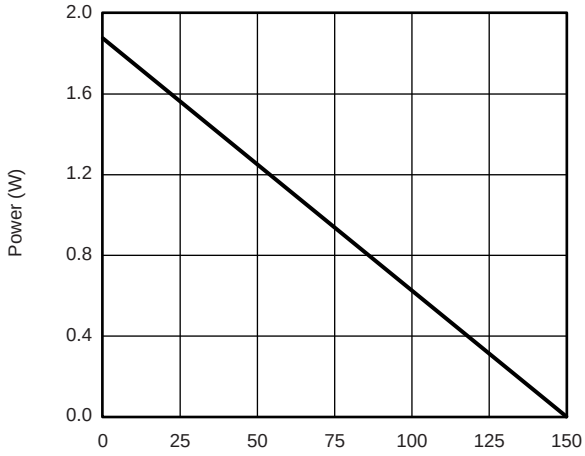
TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



Current Derating*



Power, Junction-to-Case

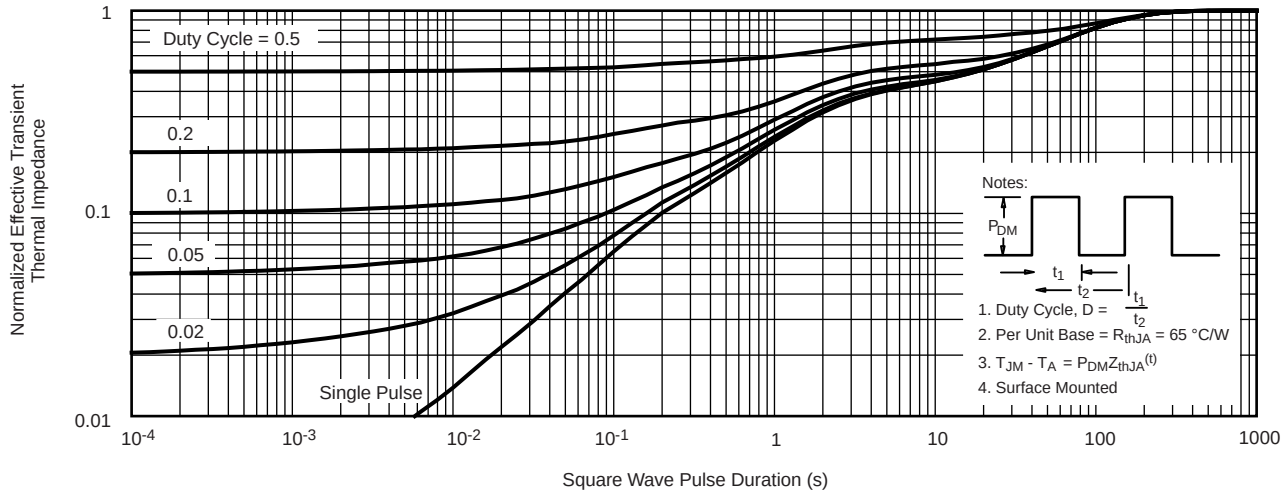


Power, Junction-to-Ambient

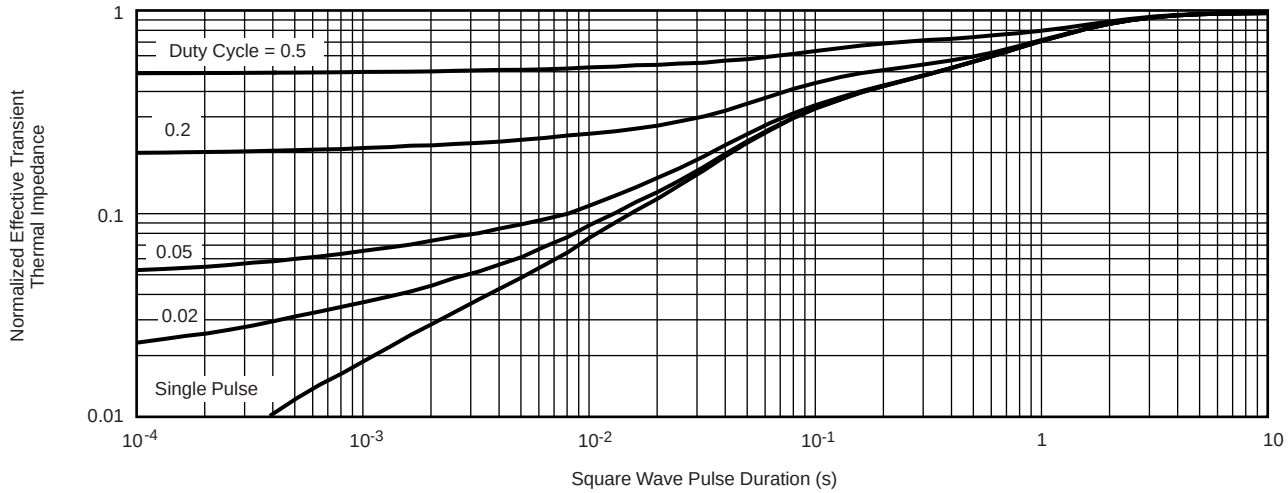
* The power dissipation P_D is based on $T_{J(max)}$ = 150 °C, using junction-to-case thermal resistance, and is more useful in settling the upper dissipation limit for cases where additional heatsinking is used. It is used to determine the current rating, when this rating falls below the package limit.

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TYPICAL CHARACTERISTICS (25 °C, unless otherwise noted)



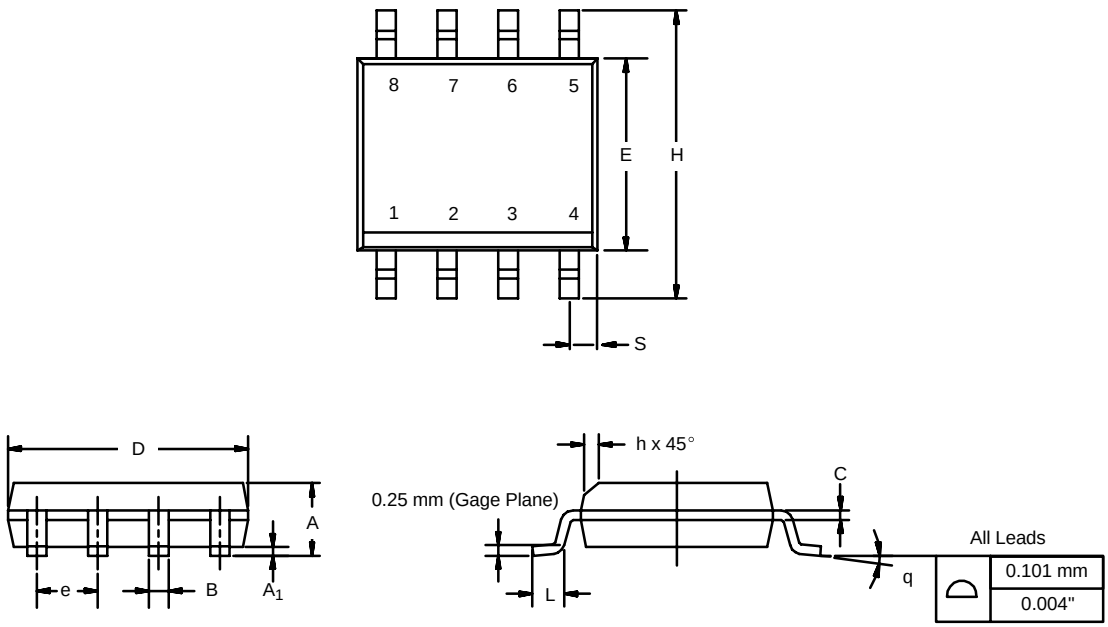
Normalized Thermal Transient Impedance, Junction-to-Ambient



Normalized Thermal Transient Impedance, Junction-to-Foot

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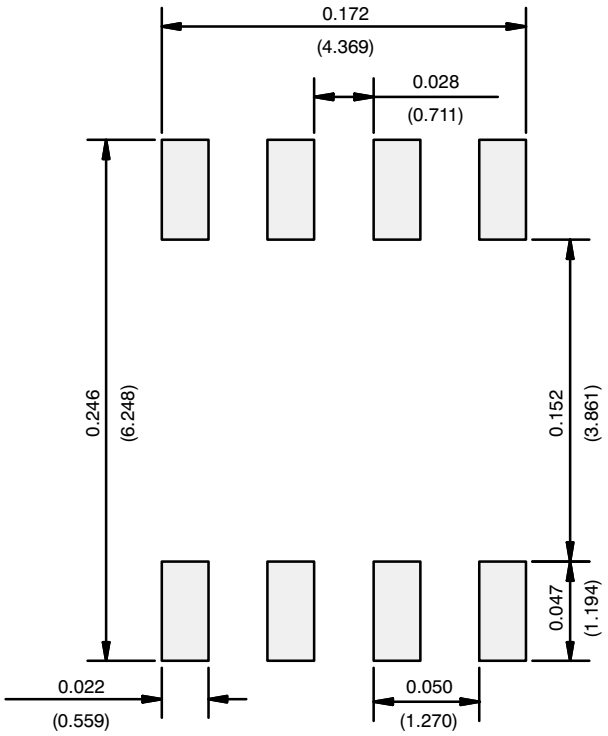
SOIC (NARROW): 8-LEAD
JEDEC Part Number: MS-012



DIM	MILLIMETERS		INCHES	
	Min	Max	Min	Max
A	1.35	1.75	0.053	0.069
A_1	0.10	0.20	0.004	0.008
B	0.35	0.51	0.014	0.020
C	0.19	0.25	0.0075	0.010
D	4.80	5.00	0.189	0.196
E	3.80	4.00	0.150	0.157
e	1.27 BSC		0.050 BSC	
H	5.80	6.20	0.228	0.244
h	0.25	0.50	0.010	0.020
L	0.50	0.93	0.020	0.037
q	0°	8°	0°	8°
S	0.44	0.64	0.018	0.026
ECN: C-06527-Rev. I, 11-Sep-06 DWG: 5498				

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RECOMMENDED MINIMUM PADS FOR SO-8



Recommended Minimum Pads
Dimensions in Inches/(mm)

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